



🎉 Congratulations! You passed!

Grade received: 100%

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CPU to GPU Algorithm Conversion Quiz

Review Learning Objectives

1. Which of the following is a classic category of non-distributed CPU algorithms used in search? 1 / 1 point

☒ Divide and Conquer

☐ Map and Reduce

☐ Attack and Triple

☐ Search and Destroy

COACH

Ready to review what you've learned before taking the quiz? I'm here to help.

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👍 Correct

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This is the correct answer. We review a search implementation of this class of algorithms.

Due Aug 21, 11:59 PM IST

Try again

2. Why is binary search algorithm like binary search not a great option for GPU? 1 / 1 point

☒ They don't efficiently use the number of available computational resources.

☐ It depends on immense scalability.

☐ Binary search only works if input data is of a size divisible by 2.

☐ Binary search requires 4 threads to work.

100%

We keep your highest score

👍 Correct

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This is the correct answer as binary search is optimized for single threaded search, though it can be implemented in a distributed manner.

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3. The simple version of distributed search shown in the slides can search millions of data points simultaneously on a AMD GPU. 1 / 1 point

☐ True

☒ False

👍 Correct

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This is correct for numerous reasons. Some are that we are learning CUDA and AMD GPUs cannot be used with CUDA and a GPU has 100s to 1000s of cores so search cannot scale on one GPU to handle millions of data points at the exact same time.

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4. Given a binary (divide and conquer) implementation of merge sort how many levels of recursion (do not include the outside "calls") to sort 16 data points? 1 / 1 point

☐ 5

☐ 4

☒ 7

☐ 0

👍 Correct

Submit your assignment

This is correct because it is one multiple of 2 more than the example, which took 5 layers, so you have one more layer added to the bottom and top of the diagram.

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